5.

Dark Matter

Gravity, the most familiar of nature's forces, offers us simultaneously the best and the least understood phenomena in nature. It took the mind of the millennium's most brilliant and influential person, Isaac Newton, to realize that gravity's mysterious "action-at-a-distance" arises from the natural effects of every bit of matter, and that the attractive force between any two objects can be described by a simple algebraic equation. It took the mind of the last century's most brilliant and influential person, Albert Einstein, to show that we can more accurately describe gravity's action-at-a-distance as a warp in the fabric of space-time, produced by any combination of matter and energy. Einstein demonstrated that Newton's theory requires some modification to describe gravity accurately—to predict, for example, how much light rays will bend when they pass by a massive object. Although Einstein's equations are fancier than Newton's, they nicely accommodate the matter that we have come to know and love. Matter that we can see, touch, feel, smell, and occasionally taste.

We don't know who's next in the genius sequence, but we've now been waiting nearly a century for somebody to tell us why the bulk of all the gravitational force that we've measured in the universe—about eighty-five percent of it—arises from substances that do not otherwise interact with "our" matter or energy. Or maybe the excess gravity doesn't come from matter and energy at all, but emanates from some other conceptual thing. In any case, we are essentially clueless. We find ourselves no closer to an answer today than we were when this "missing mass" problem was first fully analyzed in 1937 by the Swiss-American astrophysicist Fritz Zwicky. He taught at the California Institute of